Forecasting first diagnoses from synthetic EHR timelines

Research intern coding test: technical report

[AUTHOR NAME]

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Links. Code and documentation: https://github.com/Reaper8202/Research_Intern_Coding_Test. Trained model package: https://huggingface.co/Reaper8202/ehr-forecast. wandb project (training, tuning, and val-curve runs): https://wandb.ai/derek-wu820-university-of-toronto/ehr-forecast.

Summary

The submission is a fixed blend of three models trained from random weights on the 2,791 training patients: fifteen small decoder-only transformers (2 layers, width 128, about 738k parameters each; three seeds, five cross-validation folds, self-pretrained on pre-anchor event sequences and fine-tuned on the 40 labels), and two LightGBM models on code counts, one with the latest value of each numeric observation. Their logits are averaged and a per-target Platt map fitted on training out-of-fold predictions scales the result. On the official validation split (365 patients) the blend scores macro AUROC **0.767 [0.746, 0.788]** and mAP **0.258 [0.234, 0.268]**; the transformer ensemble alone scores 0.751 / 0.209 and LightGBM on counts alone 0.762 / 0.239. Test scores are the organisers'.

The transformer is the part of the system this project built and studied, and it is what the report describes. The blend was adopted at the end by a rule written before its score, because in nested cross-validation it beat both of its parts with intervals above zero. Most of the effort went into the pipeline rather than the models: a truncation rule checked against the released test files, column allowlists that keep future-bearing fields out, a prior-condition mask shared by every model, five-fold cross-validation on train for every choice, three tuning rounds with budgets committed before their scores, an independent audit of every data boundary, and two recorded val runs, both disclosed.

1. Representation

One unit of input is one clinical event token, `source:kind:code`, with its value fused in. A patient's sequence is

`[SEX:F]  hx:<older conditions> ...  [SEP]  <events, oldest first, one token each>`

Source	Token	Example
conditions, medications, care plans, allergies, devices	`src:start:code`, `src:stop:code`	`med:start:310798`
procedures, immunizations	`proc:code`, `imm:code`	`proc:430193006`
numeric observations	`obs:code:unit:q<bin>`	`obs:8480-6:mm[Hg]:q7`
text observations	`obs:code:<normalized value>`	`obs:72166-2:never_smoker`

Source	Token	Example
imaging studies, encounters	`img:modality|site`, `enc:class:code`	`enc:wellness:162673000`
genotype (D34)	`[DNA]`, `dna:<site>:<alleles>`	one token per carried variant
brain MRI (D34)	`[IMG]`	embedding replaced by a 3-D CNN encoding

Fields and modalities. Every CSV table is read through a column allowlist. Dropped on purpose: death date, lifetime healthcare expenses and coverage, marital status, addresses, all cost and payer columns, medication dispense counts and total cost (in the released test files 873 of 1,049 medications that continue past the anchor have more dispenses than months elapsed, so they leak future refills), and device UDIs (they encode manufacture and expiry dates). Race and ethnicity are not used (D15; a LightGBM check showed they cost no measurable score). Three observation codes, DALY, QALY, and QOLS, are removed (D36): they are simulator scores recorded once a year from 2011, their counts predict an anchor before 2015 with AUROC 0.999, and they tell the model how the anchor was constructed rather than anything clinical. Removing them cost about 0.014 CV AUROC in the transformer and 0.005 in the final blend, and probably costs test score, since the test files contain them.

Genomics and the brain-MRI DICOM volumes are included at my request after an earlier decision to leave them out. Coverage is thin: genotype files exist for 396 training and 2 test patients, and a pre-anchor scan for 200 training and 28 test patients. The scan enters as one token only when its study date is before the anchor day. Measured in CV they add nothing (+0.0005 AUROC; on the 541 training patients whose inputs actually change, -0.003 [-0.020, +0.015]). The volumes are near-identical simulated brains (block-mean correlation 0.995 to 0.999 between patients). The audit (section 3) found that having a genotype file or a scan is itself correlated with outcomes in this dataset, so the presence tokens could act as a proxy; the CV result says the model does not use them, and only 2 test patients have genotypes.

Continuous values. Each numeric observation is binned into quantiles fitted per code and unit on the fold's fit patients only (up to 10 bins, merged where boundaries coincide), and the bin index is fused into the token. Units stay separate; no unit conversions or clinical thresholds come from outside. A value token needs at least five fit patients, otherwise it maps to `obs:code:other_value`. A LightGBM check compared continuous latest values against their bins and found no material loss (0.243 against 0.244 mAP), so the binned representation stayed. The LightGBM part of the blend uses the continuous latest value directly, with a missing flag and the log of days since the measurement.

Time between events. Every token carries the patient's age at that event, to the day, as `age/100` plus sine and cosine terms over eight periods from 0.5 to 200 years, mapped by a learned linear layer and added to the token embedding. The model can therefore read the gap between any two tokens. Rotary position encoding gives token order. The prediction head also receives the exact time from the last event to the anchor, age at anchor, sex, and a missing-history flag. Explicit gap tokens were planned as an experiment and not reached within the budget.

Context and summary. Sequences are capped at 1,024 tokens. The most recent complete event days are kept; conditions first diagnosed before the kept window become `hx:` summary tokens (up to 128) at the front, so an old diagnosis is never lost. Under the final tokenizer about 4.6% of training patients are truncated, but they are the densest histories: 15 to 18% of all event tokens fall outside

the window, mostly medication refills and encounters. Exact refills (a stop and a restart of the same drug on the same day; 160,150 of them in fold 0) are merged first. Contexts of 512 and 2,048 were screened on fold 0 and did not help.

Leakage controls. Train and val timelines are cut with the same rules the released test files follow: keep rows dated before the anchor day, blank any `STOP` on or after it. The rule was chosen among four candidates by matching the distribution of the gap from last event to anchor in the test files (D1). The pipeline stops if any table holds a date on or after an anchor. A test adds a future record and checks that every prediction is unchanged. Labels come from the full record for train and val patients; the prediction interface never sees them. The vocabulary, bins, folds, self-pretraining windows, LightGBM vocabularies and value medians, stacking weights, and Platt maps are all fitted inside each fold on fit patients only.

2. Model and training

Architecture. Decoder-only transformer with pre-LayerNorm blocks, causal multi-head attention, RoPE, GELU MLP (4x width), token embedding plus the age projection: 2 layers, width 128, 4 heads, vocabulary about 1,930 tokens, 738k parameters in total. The pretraining head predicts the next token with weights tied to the embedding; input-only symbols are masked out of the output distribution. The prediction head concatenates the last clinical-event state with the mean state over all clinical events (the `last_mean` readout adopted in round 2) and the four head features, applies dropout, and maps to 40 logits with independent sigmoids. The MRI encoder is a four-layer 3-D CNN (81k parameters) from random weights on a 64-cubed volume.

Objectives. Self-pretraining: next-token cross-entropy over clinical event positions, on each fit patient's history strictly before the real anchor, one random full window per patient per epoch; target-diagnosis tokens weigh 5 in the loss (round 2). Fine-tuning: unweighted binary cross-entropy over the unmasked patient-target cells (a target the patient already has is masked out of the loss and set to probability 0 at prediction), normalised by the cell count across gradient-accumulation micro-batches; one extra supervised example per fit patient per epoch from an earlier anchor five years back (round 2).

Hyperparameters (`configs/experiments/v2_c1.toml`): AdamW. Pretraining: lr 0.001, betas (0.9, 0.95), weight decay 0.1, 10% warm-up then cosine to 0.1 of peak, batch 32 patients, 30 epochs, patience 3 on fixed held-out windows; every run reached epoch 30 with held-out loss still falling slowly (fold 0: 1.449). Fine-tuning: head lr 0.001, body lr 0.0001, weight decay 0.01, gradient clip 1.0, batch 32, dropout 0.3 in the body and head, up to 30 epochs, patience 5 on held-out masked macro AUROC; best epochs ranged from 3 to 13. One fold (pretraining plus fine-tuning) takes about 20 minutes on a 16 GB M-series Mac; the 15 submitted members took about 4.5 hours in all. Everything is float32.

Three training methods were compared on the same folds before any tuning: S (random initialisation, supervised only), H (pretrained body held fixed, head only), and F (pretrained body, full fine-tuning). F won on every fold: mean fold AUROC 0.760 against 0.744 (S) and 0.742 (H); pooled paired difference F minus S +0.020 [+0.013, +0.028]. Self-pretraining helps; a fixed body does not carry enough.

Tuning, pre-registered. Three rounds, each written to a budget file and committed before any of its scores existed, with a fixed adoption rule: mean AUROC gain at least 0.005, higher on at least 4 of 5 folds, mAP loss at most 0.005. Round 1 (11 trials) adopted dropout 0.3; lower body learning rate, weight decay 0.1, an age-sex head, 60 pretraining epochs, and contexts 512 and 2,048 failed.

Round 2 (12 trials, after removing the clock codes) adopted the combination of a pooled readout, extra anchors, and target-weighted pretraining; a count branch, per-target loss weights, and dropout 0.5 failed. Full-timeline pretraining (reading fit patients' post-anchor events) passed the rule and was **not** adopted, because the task says to truncate at the anchor. Three seeds of the winner were trained and averaged. Round 3 tested the one architecture dimension never varied, depth and width: a 4-layer, width-192 model (2.26M parameters) reached a slightly lower pretraining loss but scored 0.758 against 0.772 on the fold-0 screen, so the small model stayed. The full register with every score is in `docs/decisions.md` (D35, D36, D38) and `reports/transformer/`.

Comparison models and the blend. Alongside the transformer: an age-sex incidence table (the epidemiological floor, with and without the "already has it" flag), L2 logistic regression on log code counts, and LightGBM on counts and on counts plus latest values, each with settings chosen by the same folds (LightGBM's grid picked its simplest corner, 4 leaves and 100 trees). In training CV the transformer ensemble and LightGBM counts were within noise on AUROC (0.761 against 0.764 pooled out of fold) and LightGBM was ahead on mAP. Round 3 then tested output stacking with nested folds: for each fold, the stacker is fitted on the other four folds' out-of-fold predictions. The parameter-free mean of logits beat the transformer ensemble by +0.014 AUROC [+0.009, +0.019] and LightGBM counts by +0.011 [+0.007, +0.014] (mean fold AUROC 0.780, mAP 0.236), a learned per-target logistic stacker gained only 0.001 more, and the pre-registered rule adopted the fixed blend. Per-target Platt scaling of the blend, also nested, lowered Brier on 4 of 5 folds and macro expected calibration error from 0.013 to 0.010, with per-target AUROC unchanged by construction (a non-positive fitted slope falls back to the identity).

Curves. Training loss, learning rate, and gradient norm are logged per step, and held-out fold metrics per epoch, for all transformer runs in the wandb project above. Figure 1 shows the requested validation curves: every saved epoch checkpoint of the 15 members scored on the official val split after training, inside the frozen bundle. The epochs selected on the held-out training folds (dots) sit near but mostly before the val optimum; val would have picked a later epoch in 13 of 15 members and gained 0.008 AUROC on average. That choice was not made.

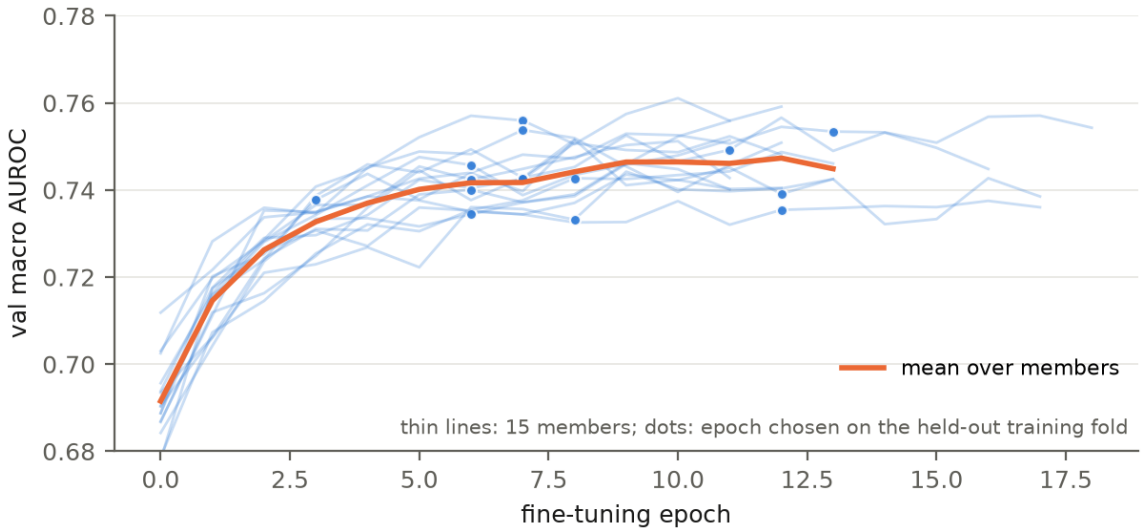


Figure 1: Official val macro AUROC of every saved fine-tuning checkpoint. Thin lines: the 15 ensemble members; dots: the epoch each member's held-out training fold selected; the orange line is the mean over members at that epoch.

Evaluation protocol. Five folds of the training split (patients never shared across folds), one out-of-fold prediction per patient from the models that did not fit on them, macro AUROC and mAP over 40 targets with the common prior mask, plus Brier score, with 95% patient-level BCa bootstrap intervals (1,000 resamples) and paired bootstraps for differences. Official val was scored by `make val-score`, which records every artifact, data, code, and lock-file hash at freeze, refuses to run on a changed tree or artifact, writes a start record before any metric and a completion record after, and refuses a second run without a documented reason. It ran twice: attempt 1 scored the transformer ensemble (0.751 / 0.209) before the blend existed; attempt 2 scored the blend after round 3 was decided on training CV. Both records are kept (`reports/val/`, `reports/val/attempts/`). The first attempt's numbers were known when round 3 was written; its rules used only training CV, and the blend's val score could not be used to change it.

3. AI workflow

Claude Code wrote most of the code, the tests, and the planning documents; I set the objective and constraints, approved or rejected each decision that touched the data or the protocol, and read every result before the next step. The division is recorded in `docs/ai_workflow.md` and the reasoning for each decision in `docs/decisions.md`.

What kept the AI's work honest: the plan was written before the code and each revision says what was known when it was made; 193 unit tests include mutation-style checks (changing < to <= in the anchor rule fails a test; a future record must leave predictions unchanged; a resumed run must match an uninterrupted one; adding held-out rows must not change the fitted vocabulary); the adoption rule was fixed before the first transformer result; every experiment block was committed before its scores; and an independent read-only audit by a second AI session checked every data boundary (truncation, allowlists, labels, tokenizer, extra anchors, pretraining windows, early stopping, modalities, LightGBM, evaluation, test split) with file and line references and found no leak.

Problems the review caught: fold-4 pretraining filled the disk with swap four times because the MPS allocator fragmented across variable-length batches (fixed by releasing cached memory each step, verified to change no computation, and 3 to 6 times faster); LightGBM and PyTorch each loading an OpenMP runtime and crashing in one process (LightGBM now runs in its own process, and the test suite pins the import order); the first bootstrap intervals using the percentile method, which for rare targets barely contained the point estimate (BCa replaced them); a preflight audit flagging its own report files as a dirty tree; and a first Platt-scaling fit that let five near-chance targets take a negative slope, which would have reversed their ranking (caught by a monotonicity test; the fit now falls back to the identity). [Derek: add anything else here, including any use of Codex CLI.]

4. Interpretation

Where the model does well. Seven targets are above 0.95 AUROC for every model: prostate neoplasm, carcinoma in situ and metastasis of the prostate, normal pregnancy, pneumonia, obesity, and the pregnancy-complication code. Each follows a visible Synthea pathway (PSA tests and rectal exams, oral contraceptives, dementia care plans, BMI observations). A leakage review of these scores found pre-anchor causes, not leaks. The transformer's clearest wins over LightGBM are the paired brain targets, silent micro-haemorrhage and abnormal gait (0.905 against 0.832 on val; the two labels are identical for every patient because one module emits both), and stroke. Figure 2 shows every target for the final blend and its two parts.

Where it does poorly, and why. Nine targets are below 0.6 for every model: whiplash, lacerations,

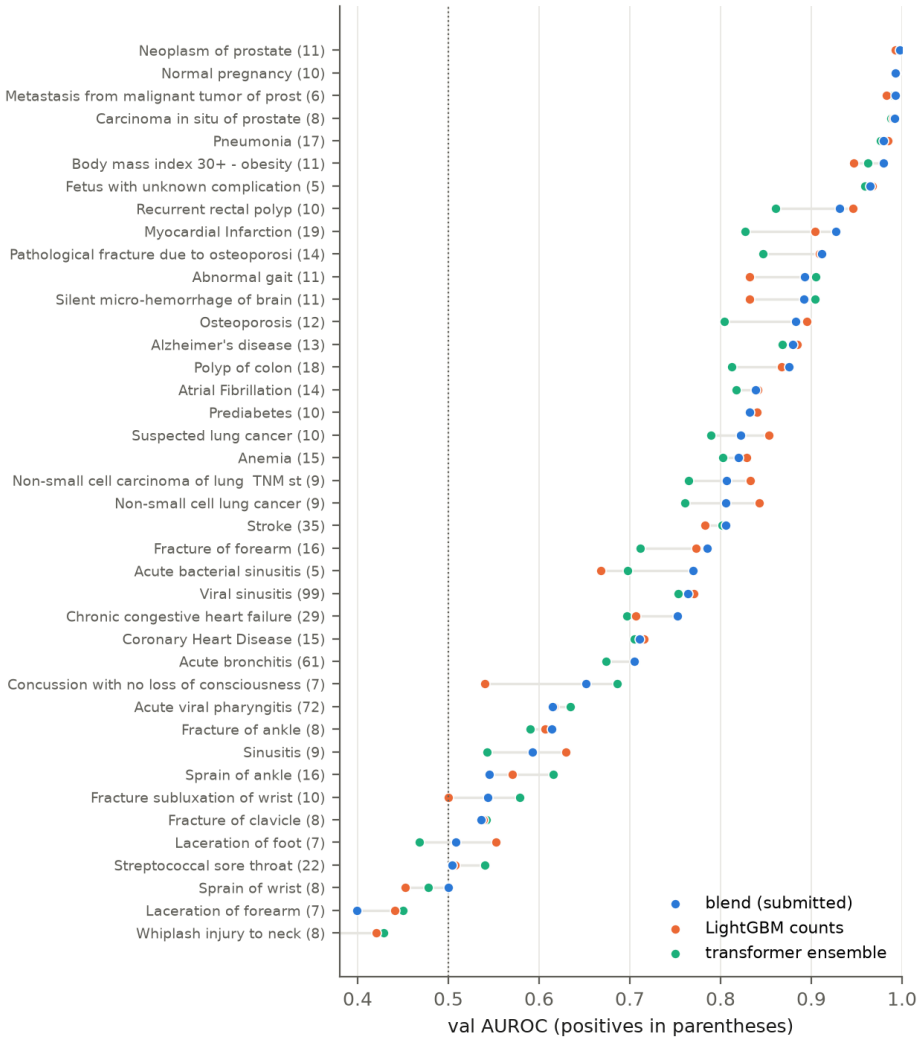


Figure 2: Official val AUROC per target for the blend, the transformer ensemble, and LightGBM on code counts, sorted by the blend. Numbers in parentheses are val positives; most targets have 5 to 20, so single-target differences are noisy.

wrist and ankle injuries, clavicle fracture, sinusitis, and strep throat. Synthea generates these acute events with little dependence on the prior record, so no history model can rank them; the age-sex table is at chance too. The transformer's largest losses to LightGBM are the count-driven conditions: osteoporosis, recurrent rectal polyp, myocardial infarction, the three lung-cancer codes, and pathological fracture, on val and in CV alike. In Synthea their risk grows with how many of a specific event has happened (screening visits, bone-density tests, nodule follow-ups), which a count feature states directly; the transformer has to infer it from a window that drops 15 to 18% of event tokens for exactly the dense histories where counts are high. The count branch tested in round 2 did not close the gap, a longer context did not help on fold 0, and that is why the blend helps: the two models fail on different targets. On val the blend is above the transformer on 29 of 40 targets and above LightGBM on 17; it recovers most of the transformer's count-driven losses (myocardial infarction 0.927 against 0.827 and 0.904, osteoporosis 0.883 against 0.804 and 0.895, pathological fracture 0.911 against 0.847 and 0.909) while keeping most of the transformer's lead on the brain pair (0.892 against 0.832). In CV the better of the two per target would reach 0.778 against 0.774 for the blend.

Anchor era. Anchors are five years before the last encounter, so an anchor before 2015 almost always means the patient died by 2020 (1,155 of 1,219 training patients), and those patients have more new diagnoses (2.37 against 1.48). Scores split by era are similar for every model, and an era signal remains in the clinical codes and age even without the clock codes (AUROC 0.865 for predicting the era). It cannot be removed without removing history, so it is reported.

Calibration. Prior-condition cells are exact zeros. The raw blend under-predicts slightly in the middle of its range (Figure 3); the Platt map corrects it. On val the blend's Brier score is 0.0365 against 0.0381 for the transformer alone.

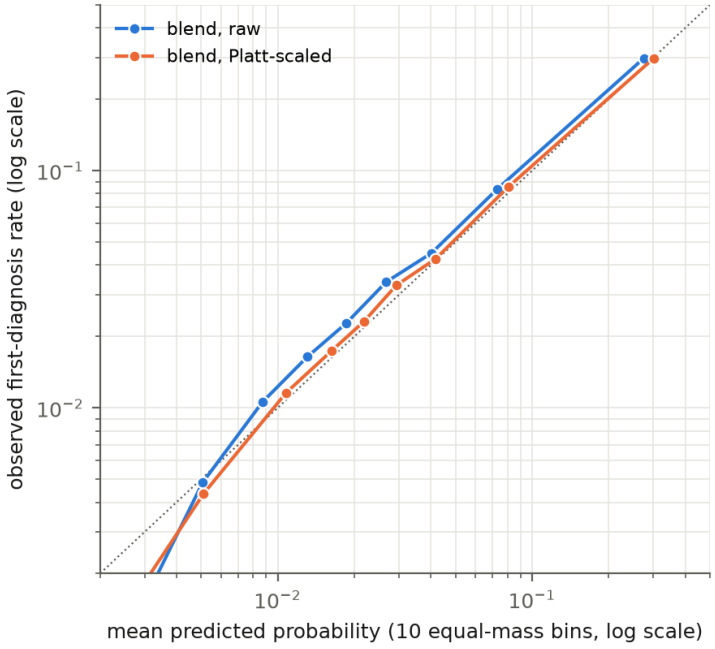


Figure 3: Reliability of the blend on training out-of-fold predictions, before and after Platt scaling, 10 equal-mass bins of all at-risk patient-target cells.

Limits. The data are synthetic and the anchor is retrospective, so nothing here is evidence about clinical use. Val is small: the macro AUROC interval is about 0.04 wide and nine targets have at most

eight positives. Every CV number for the chosen configuration is the best of 24 transformer trials plus three stackers on the same five folds, so it is optimistic; the val scores are the first check and the hidden test the second. Two seeds of CV variation (about 0.002 AUROC in the mean, up to 0.014 per fold) are comparable to several of the differences in the tuning tables. A macro AUROC of 0.80 is out of reach on this data without the clock codes: the nine near-chance targets cap the average, and even an oracle choice between the transformer and LightGBM per target reaches 0.778 in CV.

5. Results summary and reproduction

Split	Model	Macro AUROC	mAP
train CV, pooled out of fold	blend (submitted)	0.774	0.212
train CV, pooled out of fold	transformer ensemble	0.761	0.181
train CV, pooled out of fold	LightGBM counts	0.764	0.188
val (365), attempt 2	**blend (submitted)**	**0.767 [0.746, 0.788]**	**0.258 [0.234, 0.268]**
val, attempt 2	transformer ensemble	0.751	0.209
val, attempt 2	LightGBM counts	0.762	0.239
val, attempt 2	LightGBM counts + values	0.758	0.257
val, attempt 2	logistic regression, codes	0.764	0.222
val, attempt 2	age-sex incidence, at risk	0.727	0.141
val, attempt 1	transformer ensemble	0.751 [0.730, 0.773]	0.209 [0.192, 0.211]
test (358)	blend (submitted)	organisers' score	organisers' score

Paired on val, blend minus transformer ensemble: AUROC +0.016 [+0.002, +0.030], mAP +0.049 [+0.033, +0.072]; blend minus LightGBM counts: AUROC +0.005 [-0.005, +0.016], mAP +0.019 [+0.002, +0.040]. `predictions.csv` holds 358 rows by 40 code columns.

Reproduction: `make setup`, link the dataset to `data/raw`, then `make dataset`, `make modalities`, `make folds`, `make tokenize`, `make pretrain` and `make finetune` per fold and seed, `make lightgbm`, `make stacking`, `make calibration`, `make lightgbm-final`, `make freeze`, `make val-score`, `make predict`, `make submission`, `make package`. The Hugging Face package carries the 15 transformer checkpoints, 5 tokenizers, 80 LightGBM boosters with their feature specs, and the Platt weights; `python -m ehrforecast.hf_package verify` rebuilds `predictions.csv` from it on CPU and reproduced it exactly, in the development environment and in a fresh virtual environment built from `requirements-lock.txt`.